

SAMSUNG PRISM

GUIDE

Smart Guided Troubleshooting Engine

Samsung PRISM Generative AI Hackathon · 3rd Edition 2026–27

Theme ID: 02 — Smart Guided Troubleshooting Engine

Team Name: Syndicates

College: SRM Institute of Science & Technology, KTR, Chennai

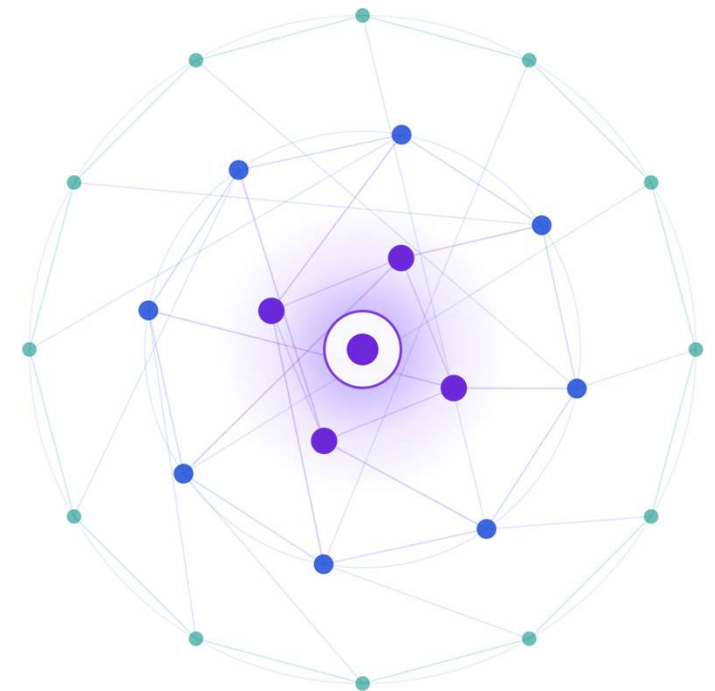
Members

Member 1 — Priyanshu Raman & Email :- pr5096@srmist.edu.in

Member 2 — Ayush Verma & Email :- av5826@srmist.edu.in

Member 3 — Harsh Vardhan Chaudhary & Email :- hc4096@srmist.edu.in

GitHub: github.com/priyanshu150106/Guide---Smart-Guided-Troubleshooting-Engine



From vague complaints to validated, one-tap troubleshooting.

Vague Complaint



Understand



Troubleshoot



Guide

Theme

THEME 02 · Smart Guided Troubleshooting Engine

THE PROBLEM

Customers describe technical issues in natural, vague language:

“My screen flickers and my battery dies really fast.”

BUT SUPPORT SYSTEMS NEED

Technical intent



Troubleshooting
procedure



Actionable
settings

TODAY'S SUPPORT JOURNEY

Customer Complaint



Manual Interpretation



Find Troubleshooting
Steps



Explain to Customer



Customer Searches
Settings

GUIDE'S VISION

Vague Complaint



AI Understanding



Structured Diagnosis



Validated Actions



One-Tap Settings

GUIDE reduces the gap between understanding a problem and actually fixing it.

Existing Solutions & Gaps

Most solutions stop at the answer. GUIDE delivers the action.

1 Traditional Support

Customer → Agent → Manual Troubleshooting

GAP

- Time-consuming
- Depends heavily on agent expertise
- Repetitive troubleshooting
- Inconsistent responses

2 Search-Based Help

Customer → Search → Support Article → Find Settings

GAP

- User must interpret the solution
- Long documentation
- Multiple navigation steps
- No personalized action flow

3 Generic AI Chatbots

Customer → LLM → Natural-language answer

GAP

- Can produce generic instructions
- May generate irrelevant steps
- Navigation links can be unreliable
- Reasoning and execution are tightly coupled

THE MISSING LAYER

Understand → Decide → Execute

Most solutions stop at: “Here’s what you should do.”

GUIDE goes further: “Here’s what you should do — and here’s the action that takes you there.”

TRADITIONAL AI

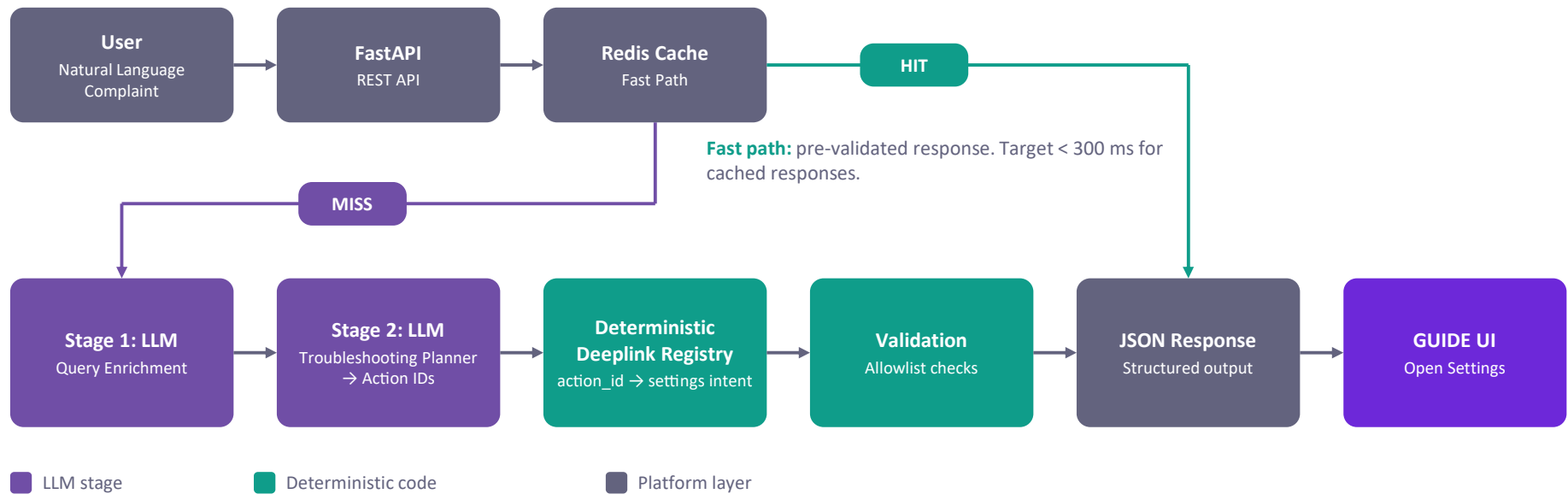


GUIDE



Our Solutions & Architecture Diagram

GUIDE — AI + Deterministic Action Layer



KEY ARCHITECTURAL PRINCIPLE

LLM decides **WHAT** action is needed.

Deterministic code decides **WHERE** the action leads.

Demo & Product Walkthrough

From Complaint → Guided Action

DEMO SCENARIO

Device: Galaxy S24 OS: Android 15

"My screen keeps flickering and my battery dies really fast."

GUIDE DETECTS



Display Flickering



Rapid Battery Drain

GUIDE GENERATES

1

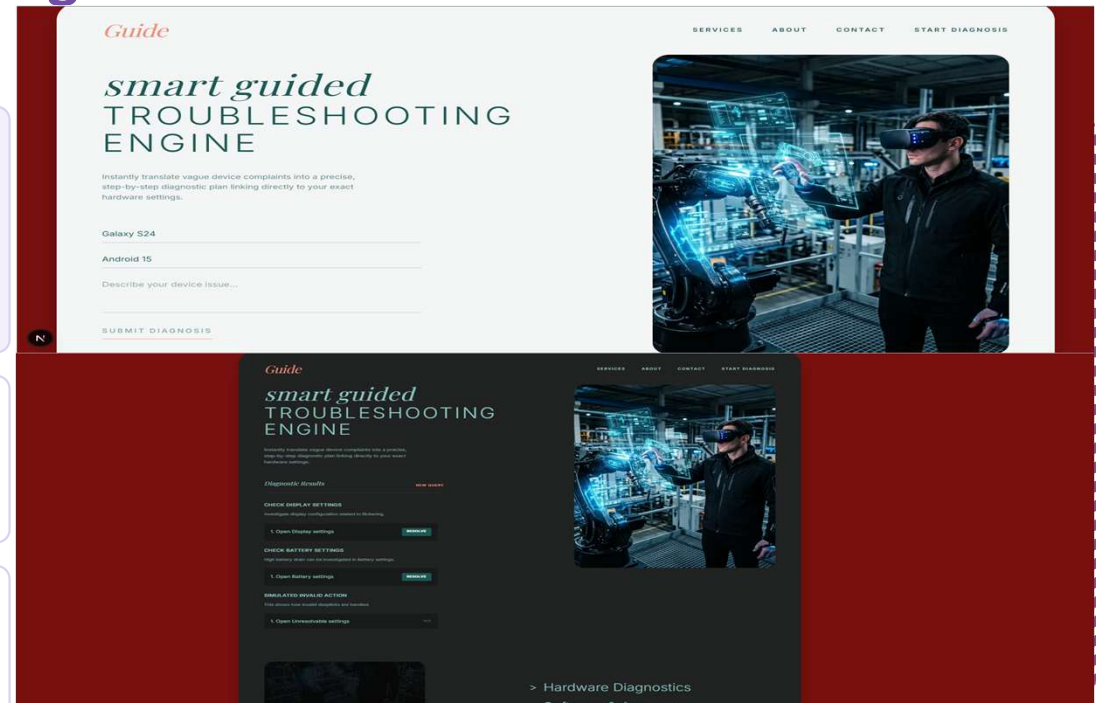
Check Display Settings

Identify relevant display configuration
Provide the corresponding Settings action

2

Check Battery Settings

Investigate battery consumption
Provide the corresponding Settings action



1. Input screen

2. AI-generated troubleshooting result

3. Action / Settings button

Describe Problem →

Diagnose →

Detected Issues →

Ordered Actions →

Open Settings

Don't just tell users what to do. Guide them to where they can do it.

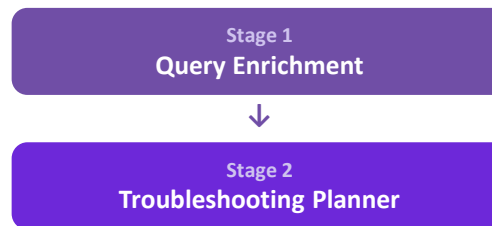
Tools and tech stack used

Technology Stack

● AI / Intelligence

- Two-stage LLM orchestration
- Query enrichment
- Structured troubleshooting planning
- Strict JSON / Pydantic validation
- Multi-provider LLM architecture

● Intelligence Layer



● Backend

- Python
- FastAPI
- Pydantic v2
- Uvicorn

● Data & Performance

- Redis — fast-path caching
- PostgreSQL infrastructure
- Telemetry
- Cost estimation
- Evaluation framework

● Frontend

- Next.js
- React
- TypeScript
- Tailwind CSS

● DevOps

- Docker
- Docker Compose
- GitHub

Architecture principle: **LLM reasoning + deterministic execution + reusable mappings**

Impact & Use case

Impact & Use Cases

01 Customer Support

Convert vague customer complaints into structured troubleshooting workflows.

IMPACT

- Reduce repetitive agent effort
- Improve consistency
- Accelerate first-step resolution

02 Self-Service Support

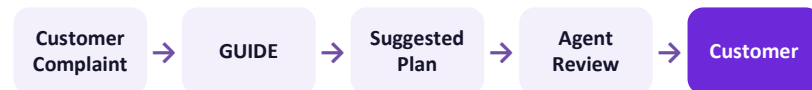
Give users actionable guidance without requiring them to search through support documentation.

IMPACT

- Lower navigation friction
- Faster access to relevant Settings
- More intuitive troubleshooting

03 Agent Assist

GUIDE can work alongside support agents.



04 Scalable Troubleshooting

Reusable action mappings allow many scenarios to share the same underlying actions.



GUIDE transforms troubleshooting from a knowledge-search problem into an action-execution workflow.

Innovation highlights, results and limitations

Innovation highlights, honest prototype status, and known limitations

1 Two-Stage LLM Architecture

Understand first → troubleshoot second

Separates intent understanding from action planning.

Intent



Planning

2 Deterministic Action Layer

The LLM returns **action_id** rather than arbitrary navigation links.

The registry resolves:

```
action_id → validated  
Android Settings intent
```

3 Semantic / Normalized Caching

"My battery dies fast"

"My battery drains quickly"

"Battery finishes too soon"



rapid_battery_drain

4 Reusable Mapping

The same action can support many scenarios.

Many Scenarios



Reusable Actions



Central Registry

Current prototype results — Verified

- ✓ Python source compilation
- ✓ Structured Pydantic contracts
- ✓ Two-stage orchestration implementation
- ✓ Deterministic action/deeplink registry
- ✓ Cache architecture
- ✓ GUIDE frontend / demo mode
- ✓ Security and repository validation

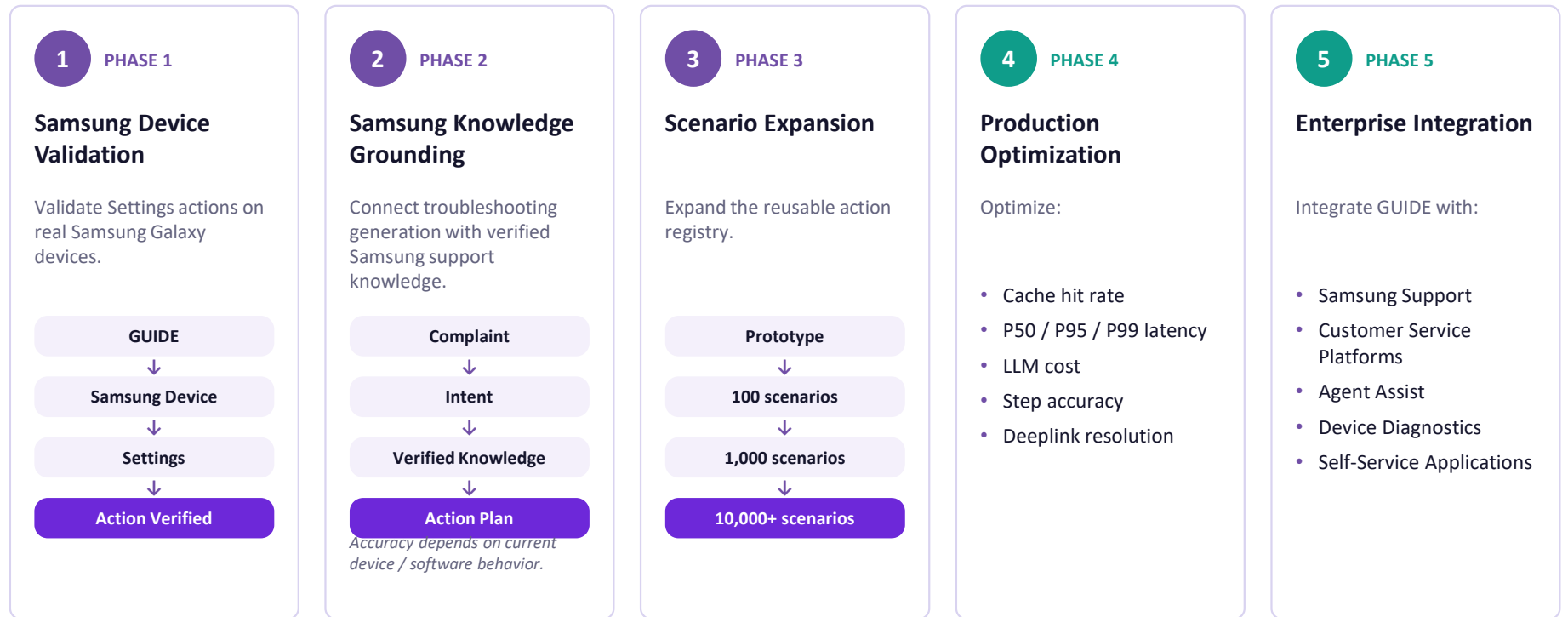
Runtime validation status

- Full Docker/API execution was not completed in the development environment.
- Physical Samsung hardware deeplink validation remains pending.

Target: < 300 ms for pre-validated cached responses

What's next

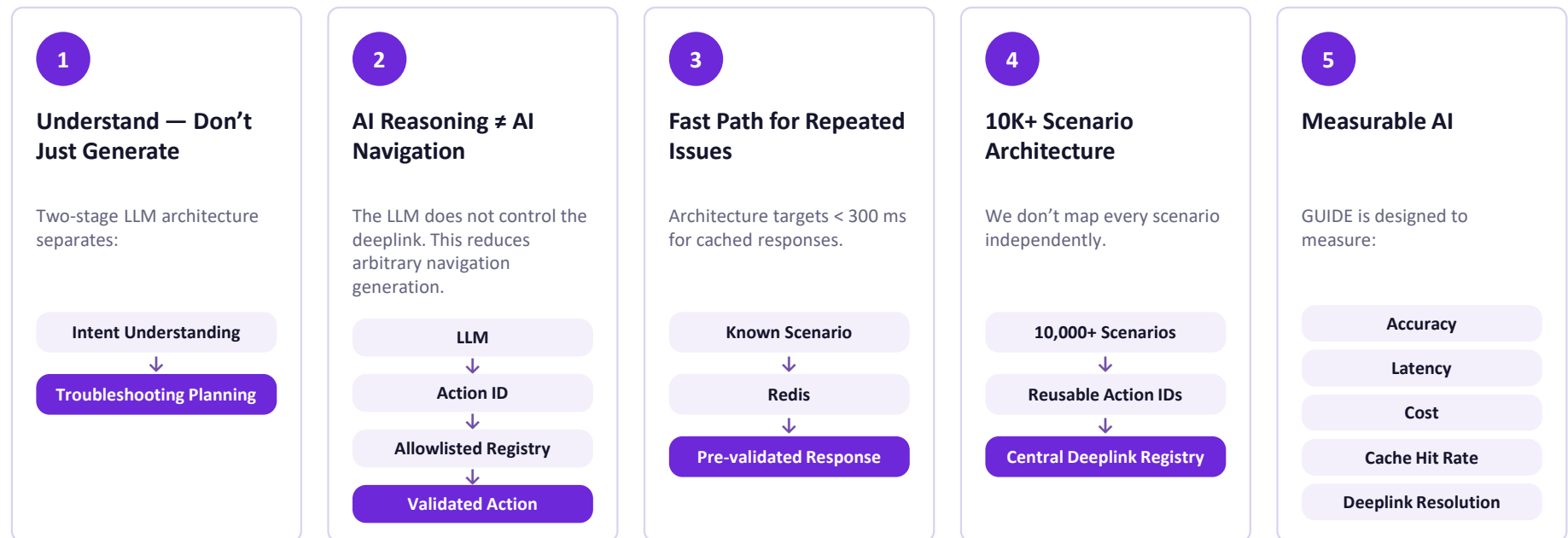
From Prototype → Production



From a hackathon prototype to an intelligent support execution layer.

Brownie points slide(differentiation)

Why GUIDE is Different



THE KEY DIFFERENTIATOR

“GUIDE is not another troubleshooting chatbot. It is an action-oriented troubleshooting engine.”

Checklist- Updated on Public GitHub

Submission Checklist

Requirement	Status
Public GitHub Repository	✓ Done
Working Prototype Code	✓ Done
README with Setup Instructions	✓ Done
Architecture Documentation	✓ Done
API Examples	✓ Done
Demo Video	✓ Done
Presentation	✓ Done
.env / secrets excluded	✓ Done
GUIDE branding finalized	✓ Done

GITHUB

[Guide — GitHub Repository](#)

github.com/priyanshu150106/Guide---Smart-Guided-Troubleshooting-Engine

DEMO VIDEO

https://drive.google.com/file/d/1wR_jI8MRMK0pl2tYJ8662JHiU56ZbCt/view?usp=sharing

Thank you

GUIDE

Smart Guided Troubleshooting Engine

**From vague complaints
to validated actions
to one-tap guidance.**

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College: SRM Institute of Science & Technology, Kattankulathur

Contact: pr5096@srmist.edu.in

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India

